

Strategies for Strategy (MSTR)

Boon Chuan Lim

Independent Researcher, Singapore

ORCID 0009-0005-8477-9393 · boonchuan@singapore.to

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Abstract

Bitcoin-treasury firms fund a non-income-producing reserve with perpetual preferred stock carrying hard cash coupons. I formalize the resulting zero-carry liability problem and, using Strategy Inc. as the worked example (\$10.0B preferred liquidation preference, 762,099 BTC, Q1 2026), order the issuer financing set by reflexive cost. A reduced-form model with one latent state variable, the perceived forced-seller probability, shows premium compression and variable-rate reset compound rather than offset, producing a low critical threshold beyond which accretive funding cannot cover the coupon. The binding constraint is coverage, not solvency. I set out the model's testable cross-asset implications: a single latent stress factor should drive preferred spreads, the equity premium, and perpetual funding with opposing signs and should lead realized issuance.

Keywords: digital-asset treasury, perpetual preferred stock, capital structure, reflexivity

JEL: G32, G35, G12, G23

1 Introduction

A new class of public company raises capital for the sole purpose of accumulating bitcoin on its balance sheet. The model works while the company funds purchases through instruments that impose no near-term cash claim — common equity issued above net asset value, and low- or zero-coupon convertible notes. As the stack matures, however, issuers turn to perpetual preferred securities that carry *cash* dividend obligations. This converts a no-obligation capital structure into one with hard, recurring claims that must be met from a balance sheet whose principal asset yields nothing.

This paper studies the resulting problem, which I call the **zero-carry liability problem**: the servicing of a fixed cash coupon out of a reserve asset that generates no cash flow.¹ The term refers to the bitcoin reserve specifically, not to the consolidated company. The issuer does operate a legacy enterprise-software business — Q1 2026 revenue was \$124.3M — but that operating cash flow is small relative to the ~\$1.05B annual preferred coupon, and management states it does not expect software operations to cover the firm's dividend and interest obligations; the coupon is therefore serviced from capital markets or asset monetization, not from operations. The contribution is threefold. First, I document the institutional structure of the preferred stack and show that its deferral optionality is asymmetric across tranches. Second, I formalize the issuer's feasible funding set and order it by **reflexive cost** — the degree to which exercising an action degrades the issuer's own future cost of capital. Third, I show that the ordering is the output of a fixed-point problem in which two adverse channels compound, producing a critical threshold beyond which only value-destructive actions remain. Strategy Inc. serves as the worked example because it is the largest and most instrument-diverse issuer; the framework applies to any bitcoin-treasury company.

The headline finding inverts the usual framing. Commentary focuses on the issuer's large unrealized loss and its distance from insolvency. I show that the binding constraint is neither: it is **coverage** — whether accretive issuance can meet the coupon — and an issuer can be far from insolvent yet sit beyond the point at which the only remaining moves destroy value.

The paper connects three literatures. The reflexive mechanism — in which the act of financing degrades the terms on which future financing is available — is the formal counterpart of Soros's (1987) reflexivity and of the feedback in DeMarzo and He (2021), where investors anticipate future issuance and widen spreads enough to offset its benefit. The asymmetry between raising and reducing a fixed claim, and the bias toward asset sales when forced to deleverage, are the preferred-stock analogue of the leverage ratchet effect of Admati, DeMarzo, Hellwig, and Pfleiderer (2018). And the departure point is the frictionless benchmark of Modigliani and Miller (1958): here the financing claim is serviced not from asset cash flow but from a reserve that yields nothing, so capital-structure choice is not irrelevant but existential. The contribution is to combine these into a single coverage condition for a non-yielding-reserve issuer and to show the threshold is reached well before insolvency.

¹Aggregate liquidation preference of \$10.005B and 98.9M preferred shares outstanding, and the per-series splits, are all exact (Form 10-Q, 31 Mar 2026, Note 9, Redeemable Preferred Stock). The five per-series liquidation preferences sum to the \$10.005B balance-sheet total to the dollar. The STRC tranche grew further after quarter-end (to roughly \$8.54B notional by 26 Apr 2026); the paper uses the 31 Mar 2026 figures throughout.

2 Institutional background: the preferred stack

As of 31 March 2026 the issuer had five series of perpetual preferred stock outstanding, each with a \$100 stated value, listed in order of seniority below. All sit junior to the convertible notes and senior to the common. The series differ in coupon, payment frequency, convertibility, and — critically for what follows — in the cost of skipping a payment.

Series	Coupon	Features	Deferral mechanics (the key asymmetry)
STRF	10% fixed	Most senior; cumulative; cash-only	Penal: unpaid dividends accumulate and compound at an increasing rate (the filing references a stepped “Compounded STRF Dividend Rate”). Sustained non-payment also triggers board-representation rights for holders.
STRC	Variable (11.5%)	Largest tranche; monthly cash; par-targeting	Rate resets monthly to hold price near \$100 — so under price stress the rate resets <i>up</i> , raising the cash bill exactly when cash is tightest.
STRE	10% (EUR)	Senior; cumulative; euro-denominated	Cumulative; unpaid dividends compound (the filing references a “Compounded STRE Dividend Rate”). Euro-denominated, so payment also carries USD/EUR currency exposure.
STRK	8% fixed	Convertible 0.1 MSTR; cumulative	Cumulative (arrears owed) but lowest coupon; sustained non-payment triggers board-representation rights. The conversion feature means the claim can, as a matter of economics rather than a contractual term, be diluted away through conversion if MSTR re-rates upward.
STRD	10% fixed	Most junior; non-cumulative	Non-cumulative: if the board does not declare, holders receive nothing and the payment is <i>never owed</i> . The only freely skippable coupon in the stack.

The asymmetry is the point: the cheapest coupon to skip (STRD) is the smallest and most junior, while the senior cumulative tranches impose a penal cost on deferral. “Just stop paying” is therefore available only at the margin.

3 Data: the cash bill

Hard figures (10-Q, 31 Mar 2026): total preferred liquidation preference \$10.005B across 98.9M shares; Q1 preferred dividends paid \$229.5M and declared \$250.6M; 762,099 BTC at \$51.6B fair value. The forward coupon run-rate is \$1.05B per year at a weighted coupon of 10.5%. The per-series figures below are exact, from the Note 9 schedule (the per-series liquidation preferences sum to the \$10.005B balance-sheet total to the dollar).²³

Series	Liq. pref.	Coupon / type	Annual cash	Skippable?
STRC	\$5.025B	11.5% variable, monthly	\$578M	Resets up under stress
STRF	\$1.284B	10% cumulative	\$128M	Cumulative; compounds if

²This is the structural distinction from a leveraged-equity or mortgage-REIT structure, in which the financing claim is serviced from asset cash flow. Here the reserve asset produces no cash flow, so the coupon is met only from external capital or asset monetization.

³Parameters are illustrative and chosen to expose signs and the existence of a threshold, not to forecast. The qualitative results hold for any $m(\cdot)$ strictly decreasing in p and any $B(\cdot)$ non-decreasing in p with a strictly positive variable-rate component.

Series	Liq. pref.	Coupon / type	Annual cash	Skippable?
				deferred
STRK	\$1.402B	8% cumulative, convertible	\$112M	Cumulative; convertible
STRD	\$1.402B	10% non-cumulative	\$140M	Freely skippable
STRE	\$0.892B	10% EUR cumulative	\$89M	Cumulative (EUR); compounds if deferred
TOTAL	\$10.005B	(par / liquidation pref)	\$1.05B	<i>\$140M freely skippable</i>

The figures are exact (Note 9). STRC alone is 55% of the annual bill and resets upward under stress; only STRD, at 13% of the bill, is freely skippable, since its dividends are non-cumulative. The cumulative tranches (STRF, STRC, STRE, STRK) accrue if deferred, and for STRF and STRK a sustained non-payment also triggers board-seat rights for holders — making deferral on those series costly in control terms as well as cash terms.

4 The feasible funding set

Facing the coupon, the issuer has six broad actions. Listed cheapest-first by reflexive cost (formalized in Section 5), with the reserve asset preserved until the last resort:

1. **Accretive issuance.** Issue common or converts while mNAV exceeds 1. The original flywheel; no cash claim created. The only fully sustainable source.
2. **Derivative monetization.** Covered calls or collars on a slice of the stack generate cash while retaining the coins; cost is forgone convexity, not a spot print.
3. **Variable-rate absorption.** Let the par-defending reset run. Mechanically self-limiting but procyclical — it raises the bill under stress.
4. **Non-cumulative skip.** Skip the junior non-cumulative tranche (STRD). Legally costless and never owed back, but a loud distress signal; small in magnitude.
5. **Cumulative deferral.** Defer senior cumulative coupons. Arrears accumulate and compound, and sustained non-payment on STRF or STRK triggers board-representation rights for holders. Avoid unless already distressed.
6. **Spot sale.** Sell bitcoin. At a large share of supply the issuer may not be a price-taker; a large or perceived-forced sale may create adverse price impact and signal distress, potentially re-rating the asset, the equity, and the preferred yields simultaneously.

5 A reflexive-cost ordering of the funding set

This section orders the actions of Section 4 by reflexive cost and shows the ordering is the output of a fixed-point problem. The central result: the two adverse channels — premium compression and variable-rate reset — **compound** rather than offset, producing a critical threshold beyond which accretive funding cannot cover the coupon.

5.1 Setup

Let N be the net asset value of the reserve asset and $p \in [0, 1]$ the market's perceived probability that the issuer becomes a forced seller within the horizon. The mNAV premium m is decreasing in p :

$$m(p) = m_0 (1 - \lambda p), \quad \lambda > 0, \quad m_0 > 1.$$

The linear specification is deliberately conservative.⁴

Accretive issuance capacity per period rises with the premium over par:

$$K(p) = \kappa \cdot \max\{m(p) - 1, 0\} \cdot N, \quad \kappa > 0.$$

The coupon bill combines a fixed component and a variable-rate component that resets up under stress:

$$B(p) = B_f + B_v (1 + \varphi p), \quad \varphi > 0.$$

The two assumptions $m'(p) < 0$ and $B'(p) > 0$ are institutional, not modelling conveniences: the first follows from accretive issuance requiring a premium, the second from the par-defending reset mechanics of the variable-rate tranche.

5.2 Coverage and the reflexive threshold

Define coverage as accretive capacity net of the coupon bill, $C(p) = K(p) - B(p)$. When $C(p) \geq 0$ the coupon is met from accretive issuance (benign regime). When $C(p) < 0$ the issuer must use reflexively costly actions. The boundary is p^* solving $C(p^*) = 0$.

Proposition 1 (compounding channels). Coverage is strictly decreasing in p , and its derivative decomposes into two same-signed terms:

$$C'(p) = \kappa N m'(p) - B_v \varphi = -\kappa N m_0 \lambda - B_v \varphi < 0.$$

The first term is the **issuance channel** (higher p compresses the premium, shrinking capacity); the second is the **coupon channel** (higher p lifts the variable rate, enlarging the bill). Both are negative, so they *amplify*. This is the analytical core: stress does not merely raise the bill or merely close the window — it does both at once, driven by the same state variable.

Proposition 2 (existence of a finite trap). If $C(0) > 0$ and $C(1) < 0$, a unique $p^* \in (0, 1)$ exists with $C(p) < 0$ for $p > p^*$. There the feasible set collapses to deferral and spot sale; since spot sale itself raises p , the system admits a self-fulfilling forced-sale equilibrium in which anticipated forced selling brings forced selling about.

Under the linear specification the threshold has a closed form. Setting $C(p) = 0$ and solving for p gives:

$$p^* = [\kappa N(m_0 - 1) - B_f - B_v] / [\kappa N m_0 \lambda + B_v \varphi].$$

The denominator is strictly positive, so a single root exists. The economically relevant case $0 < p^* < 1$ obtains exactly when $C(0) > 0$ and $C(1) < 0$, i.e. when $\kappa N(m_0 - 1) > B_f + B_v$ (coverage is positive at $p = 0$) and $\kappa N(m_0(1 - \lambda) - 1) < B_f + B_v(1 + \varphi)$ (coverage is negative at $p = 1$). Both numerator terms that depend on the variable-rate tranche, B_v and φ , push p^* lower — the analytic counterpart of Proposition 1: a larger variable-rate tranche brings the trap closer.

⁴Q1 2026 preferred dividends were \$229.5M paid and \$250.6M declared. The forward run-rate of \$1.05B is built from the exact per-series liquidation preferences times their stated coupons (Note 9); this exceeds a naive annualization of Q1 cash paid because the variable-rate STRC tranche grew through the quarter, so its quarter-end par carries a full-year coupon the partial-period cash figure does not capture.

5.3 Calibration

Figure 1 plots $K(p)$ and $B(p)$ under an illustrative calibration anchored to the Q1 2026 balance sheet ($N \approx \$51.6\text{B}$, $B \approx \$1.05\text{B/yr}$).⁵ The premium begins above par, so coverage is positive at $p = 0$; but because the channels compound, coverage crosses zero at a *low* value — here $p^* \approx 0.10$ — well before the premium reaches par. The shaded region right of p^* is the reflexive trap.

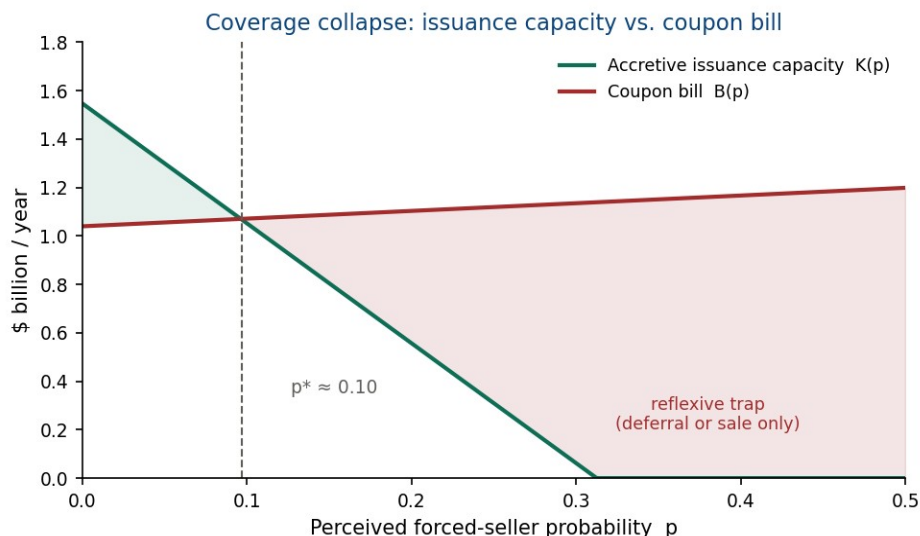


Figure 1. Accretive issuance capacity collapses below the coupon bill at a low critical forced-seller probability p^* . Both channels move coverage the same way, so the crossing occurs while the premium is still above par.

5.4 The induced ordering

The model delivers the waterfall of Section 4 as an ordering by marginal effect on p . Actions that leave p unchanged are reflexively cheap (accretive issuance, derivative monetization); actions that raise p are reflexively expensive, and convexly so by Proposition 1 (variable-rate absorption and non-cumulative skip raise p modestly; cumulative deferral and spot sale raise it sharply). The ranking is thus derived, not asserted.

5.5 What the model says that the narrative does not

First, the binding constraint is coverage, not solvency: an issuer can withstand a deep drawdown yet sit beyond p^* . **Second**, concentration in the variable-rate tranche is not neutral — it raises B_v and ϕ 's leverage on coverage, lowering p^* and shrinking the benign region. **Third**, the cheapest financial lever (derivative monetization) is in tension with the brand commitment that sustains m_0 , so even reflexively cheap actions carry a premium cost — a coupling captured by letting m_0 depend on observed hedging policy.

⁵The linear form is the conservative case. If the premium compresses convexly in p — plausible once the market begins to price a forced-seller scenario, since confidence in the equity-issuance channel can collapse faster than linearly — then $m'(p)$ steepens as p rises, lowering the critical threshold p^* and accelerating the loss of coverage. Linearity therefore understates the fragility the model identifies.

6 Testable implications

The model makes a sharp, falsifiable prediction. If the mechanism is real, the preferred-yield spread, the equity mNAV premium, and the perpetual-futures funding rate should co-move with a single latent factor — the perceived forced-seller probability p — with the **opposing signs** of Section 5.1: as p rises, spreads widen while the premium and funding fall. That common-factor structure, with those signs, is the empirical content of Proposition 1, and is in principle recoverable from a state-space model of the three daily series. A second prediction concerns direction: innovations to p should **lead**, not lag, realized accretive issuance and the derivatives footprint — perceived forced-seller risk moves first and the financing response follows. A third concerns the threshold of Proposition 2: the sensitivity of issuance to p should weaken in the high- p regime, as the accretive channel stops absorbing stress near p^* . A natural design recovers $\hat{p}$ jointly from the three observables and tests its lead–lag structure against issuance and funding; a cross-issuer version asks whether issuers with greater variable-rate concentration exhibit a lower p^* . I leave the estimation to future work.

7 Discussion

The zero-carry liability problem generalizes beyond any single issuer to the broader class of digital-asset treasury companies that finance a non-yielding reserve with cash-coupon claims. The framework clarifies when the issuance window closes, why variable-rate concentration is destabilizing, and why the reflexive trap is distinct from — and reachable well before — insolvency. The limitation is that the model is reduced-form: a micro-foundation for $m(p)$ and a formal robustness result establishing invariance to functional form are natural next steps, alongside the empirical test set out in Section 6.

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All figures are anchored to Strategy Inc. Form 10-Q for the quarter ended 31 March 2026 (aggregate preferred liquidation preference \$10.005B; 98.9M preferred shares; Q1 preferred dividends \$229.5M paid / \$250.6M declared; 762,099 BTC at \$51.6B fair value), with per-series terms and amounts taken from Note 9. The forward coupon run-rate is constructed as the sum of per-series liquidation preference times stated coupon. Calibration parameters are illustrative. This is a working research draft and not investment advice. The author holds no position disclosed herein.